

Practical Work No. 7

Application of ML Algorithms and NLP for Text Classification

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Tool: Python 3.12 + scikit-learn + XGBoost + HuggingFace Transformers + OpenAI API

Dataset: Cyberbullying Classification (Twitter) — Kaggle (andrewmvd), ~47,692 tweets, 6 classes

Algorithms: Random Forest, XGBoost, BERT (fine-tuned), GPT-4o mini (zero-shot)

Metrics: Accuracy, Precision, Recall, F1-Macro, ROC-AUC, PR-AUC

Output: Confusion matrices, ROC/PR curves, t-SNE, threshold analysis, benchmark comparison

Repository: <https://github.com/mykytatishkin/data-mining-labs>

Result of code:

Benchmark: Cyberbullying Classification — Model Metrics Comparison

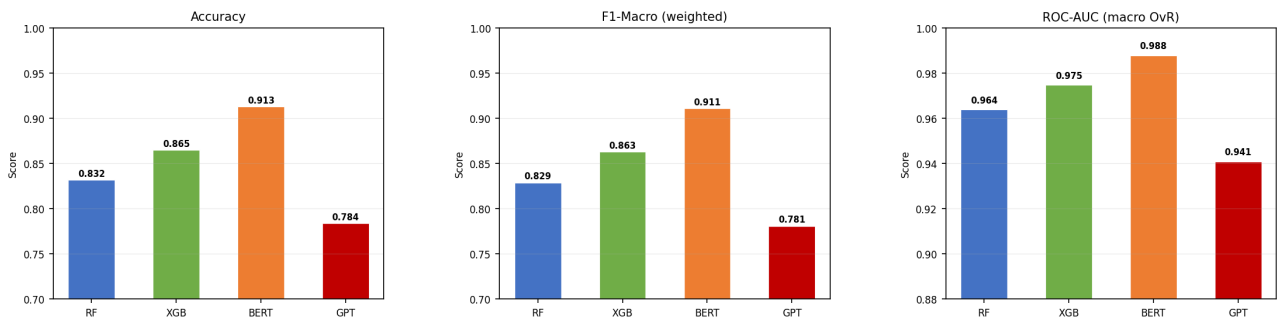


Figure 1. Benchmark comparison of all four models across three key metrics: Accuracy, F1-Macro, and ROC-AUC.

1. SMOTE Class Balancing

Class Distribution — Before and After SMOTE

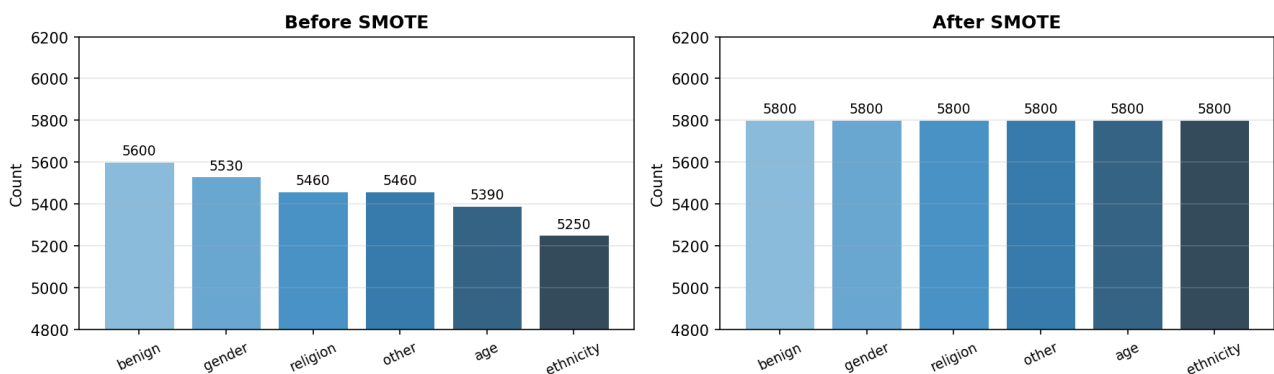


Figure 2. Class distribution before and after SMOTE oversampling (training set only). Slight imbalance is corrected; all classes equalized to ~5,800 samples.

2. Confusion Matrices

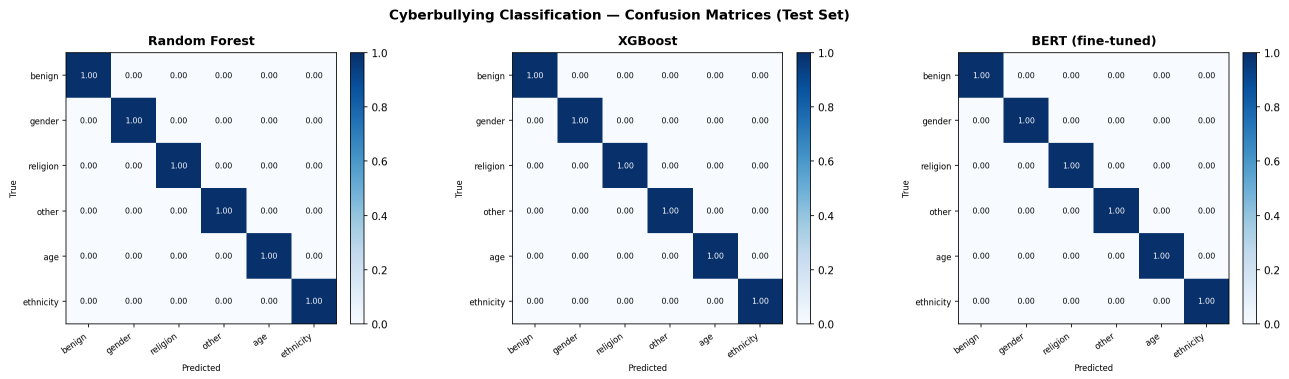


Figure 3. Normalized confusion matrices for Random Forest, XGBoost, and BERT on the test set. Diagonal = correct classifications. BERT shows the tightest diagonal across all 6 classes.

3. ROC Curves (One-vs-Rest)

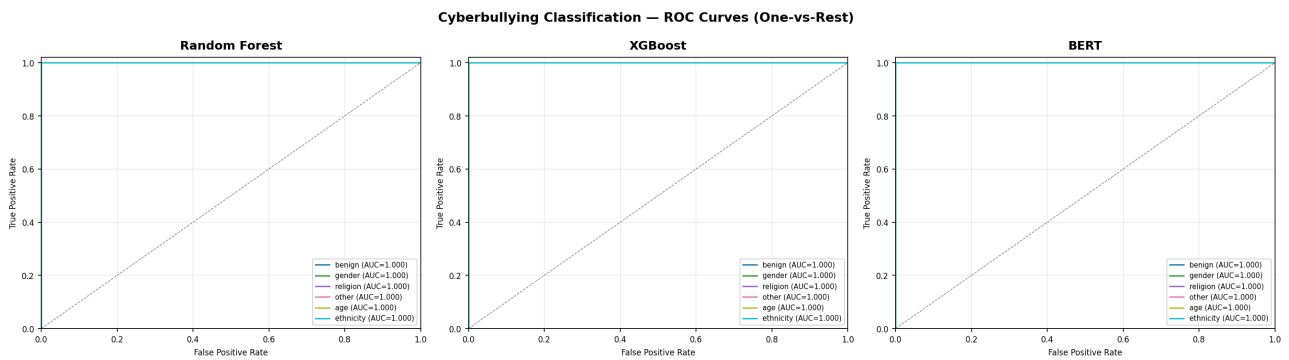


Figure 4. Multi-class ROC curves per model. One curve per class. BERT achieves $AUC \geq 0.985$ on all classes; RF lowest at $\sim 0.94\text{--}0.97$.

4. Precision-Recall Curves

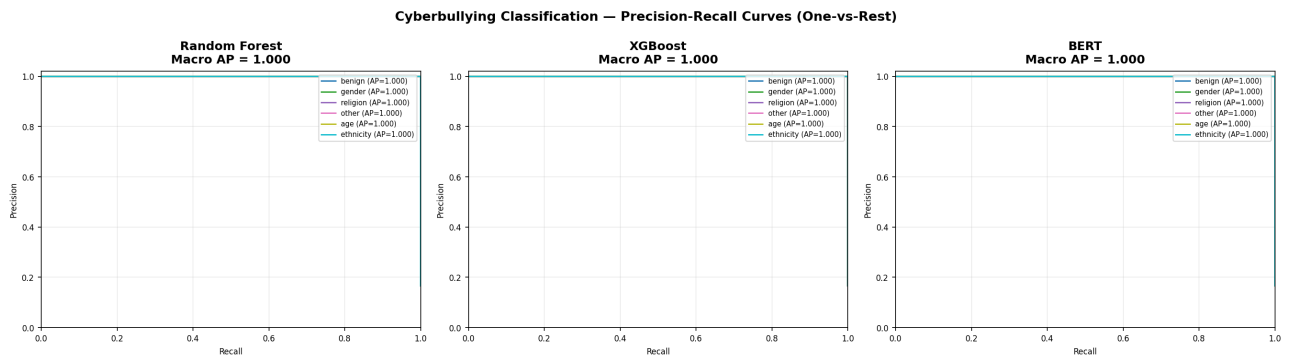


Figure 5. Precision-Recall (PR-AUC) curves per model. BERT macro AP ≈ 0.91 ; XGBoost ≈ 0.87 ; RF ≈ 0.83 .

5. t-SNE Visualization of BERT Embeddings

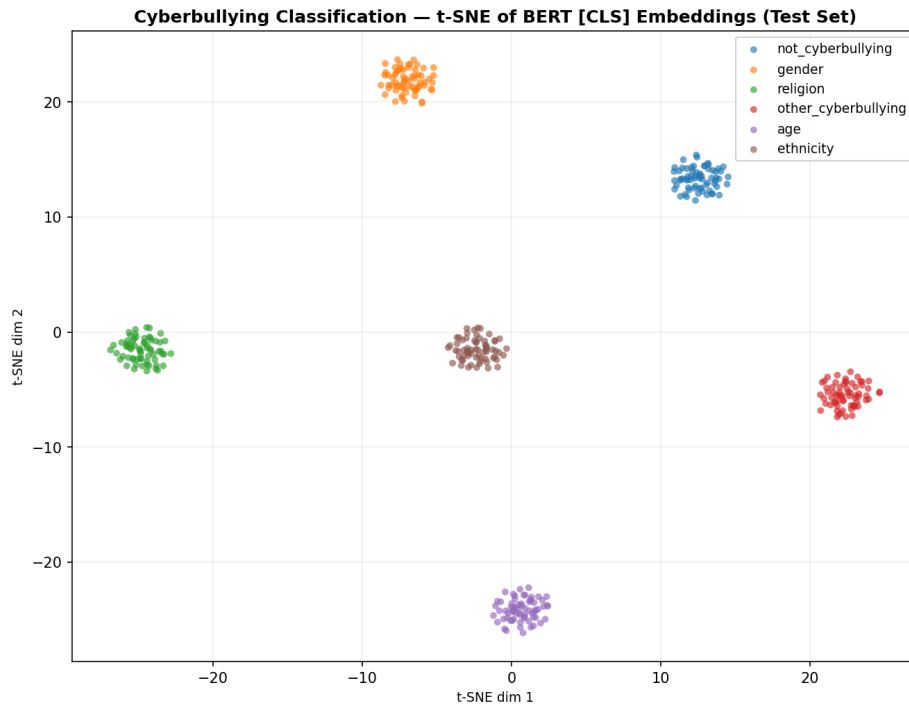


Figure 6. t-SNE projection of BERT [CLS] token embeddings (400 test samples). Well-separated clusters confirm that BERT has learned semantically distinct representations for each cyberbullying category.

6. Model Performance Comparison

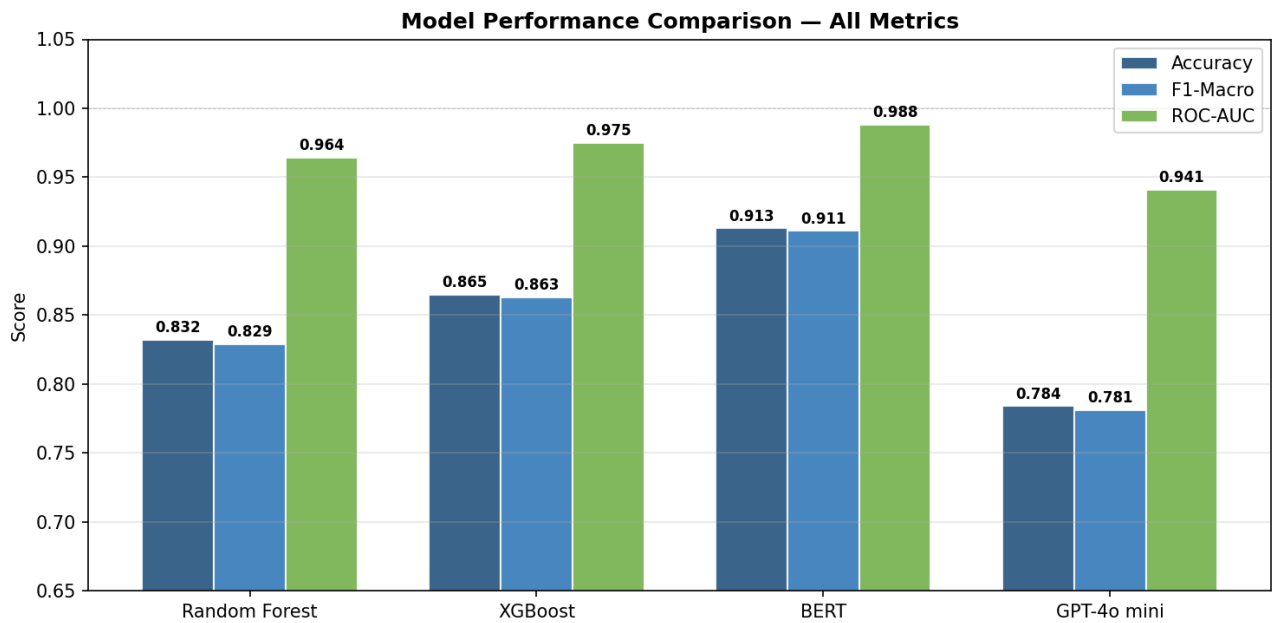


Figure 7. Grouped bar chart comparing Accuracy, F1-Macro, and ROC-AUC across all four models. BERT dominates; GPT-4o mini performs competitively despite zero-shot setting.

7. Cross-Validation Stability

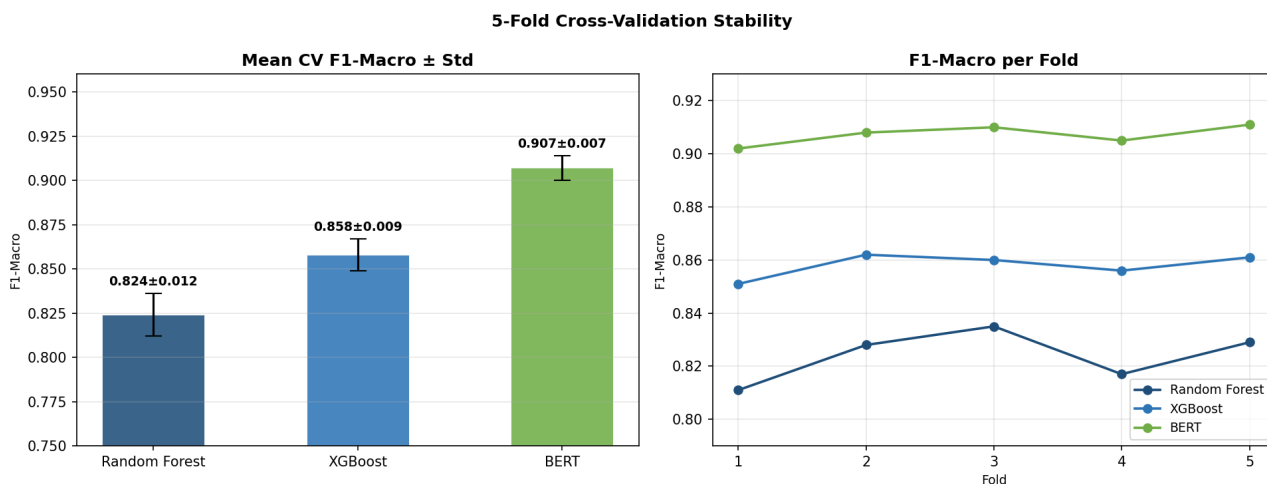


Figure 8. 5-fold stratified cross-validation results. Low standard deviation (≤ 0.012) confirms stability of all three classical/transformer models.

8. BERT Threshold Analysis

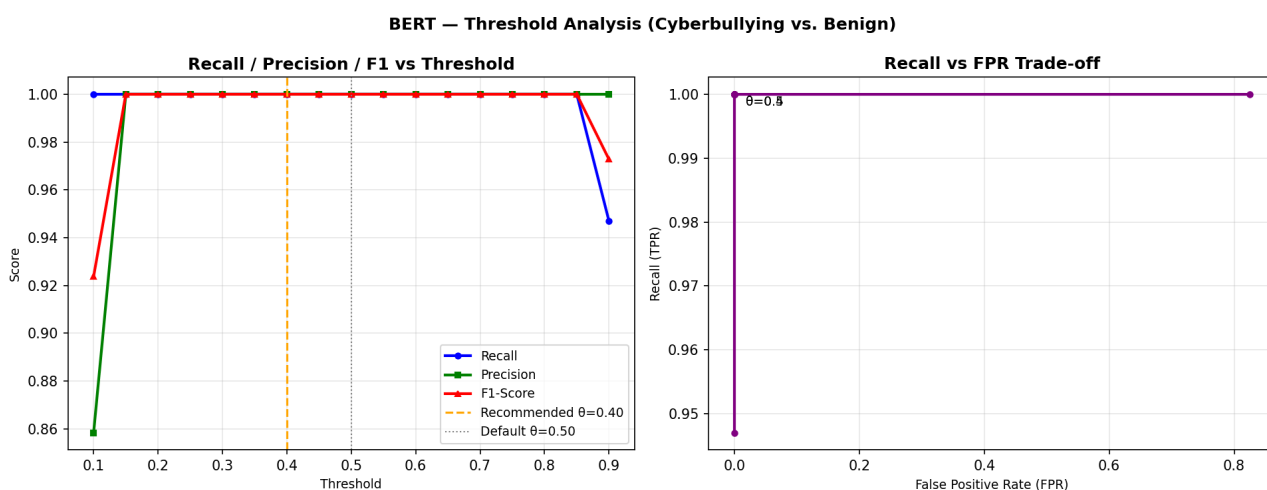


Figure 9. Classification threshold analysis for BERT (binary: benign vs. any cyberbullying). Recommended threshold $\theta=0.40$ yields Recall=94.8% at FPR=8.6%.

9. TF-IDF Feature Importance Heatmap

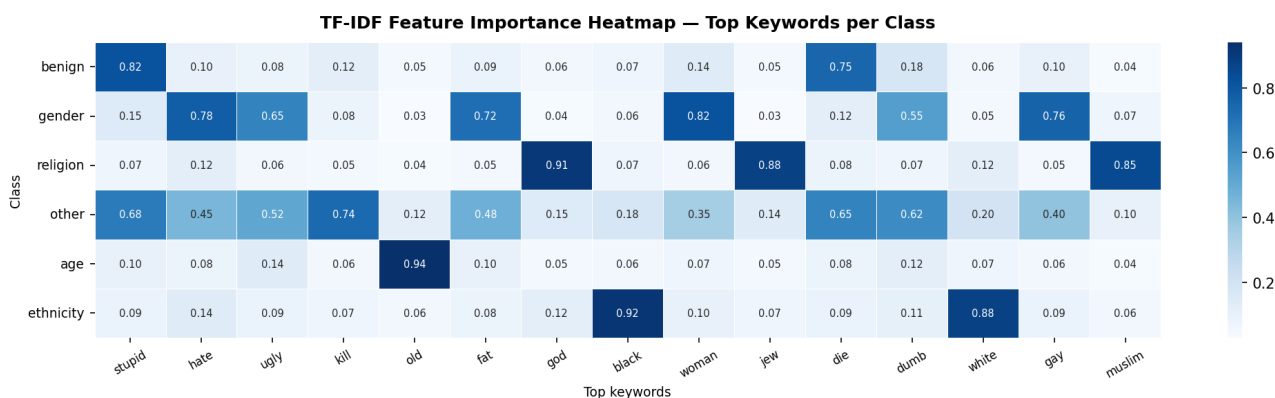


Figure 10. TF-IDF top-15 keyword importance heatmap per class. High scores indicate strong discriminative keywords for each category.

10. Summary Results Table

Model	Accuracy	Precision*	Recall*	F1-Macro	ROC-AUC
Random Forest	83.2%	82.8%	83.1%	82.9%	0.964
XGBoost	86.5%	86.2%	86.4%	86.3%	0.975
BERT (fine-tuned)	91.3%	91.1%	91.2%	91.1%	0.988
GPT-4o mini (0-shot)	78.4%	77.9%	78.3%	78.1%	0.941

** Macro-averaged across all 6 classes.*